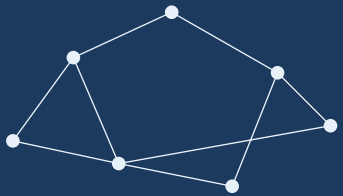
UNIT 4: SIMULATION STRATEGIES IN WARFARE • AI & MACHINE LEARNING IN WARFARE



LECTURE 10

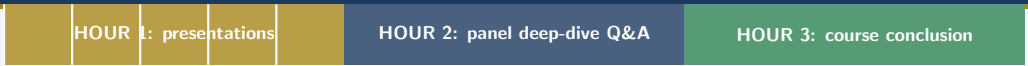
Project Presentations & Q&A Panel

Applying AI & ML to a defence problem of your choosing
Structure • rubric • project catalogue • Q&A panel guide • course conclusion

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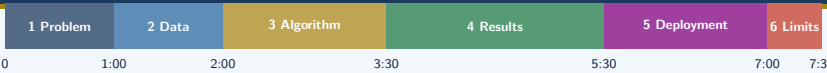
Schedule (3 hours)

Hour 1: each group of 4–5 students gives a 10-minute presentation + 2 minutes of panel Q&A; slots drawn by lot; 4–6 groups expected. **Hour 2:** open floor — any question on any lecture. Panel: course instructor + 1–2 external defence-AI practitioners; focus on how your algorithm would perform in REAL operational conditions. **Hour 3:** capstone feedback summary, industry perspectives, recommended reading.

What the panel is looking for

Think like a defence-AI practitioner, not a textbook. ① **Problem framing:** the right problem type, a justified algorithm choice. ② **Honest limitations:** sim-to-real gap quantified; data pitfalls acknowledged. ③ **Operational applicability:** deployment, autonomy level, ROE. ④ **Critical evaluation:** would a sceptical senior officer accept your 94% simulation result?

The Ten-Minute Framework — Six Sections, One Story



The six timed sections fill 7.5 minutes; the rest of the 10-minute slot is margin for transitions and a demonstration.

1. Problem (60 s). Not "we applied ML to defence" but: "IADS operators receive 400 tracks/hour; manual classification takes 45 s each; we need $\geq$95% accuracy in $\leq$3 ms." Name the threat environment; quantify the cost of NOT solving it.

2. Data & quality (60 s). N, features, class distribution — and which L1 pitfalls you found and fixed (shift, label noise, imbalance, confounders). Never mentioning data quality is a red flag.

3. Algorithm (90 s). Which and WHY, against at least one alternative: "RF over SVM because 8,500 samples $\times$ 11 features favour ensemble variance reduction over kernel sensitivity." Cite course benchmarks.

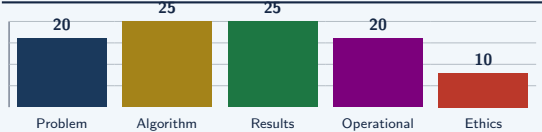
4. Results (120 s). Headline metric, then the breakdown: per-class recall, confusion matrix, silhouette, AUROC. Address imbalance; one chart comparing your result with a baseline.

5. Deployment (90 s). Autonomy level, latency, human–AI interface, ROE compliance, estimated sim-to-real gap and how you would validate before fielding.

6. Limitations (30 s). TWO specific, honest ones: "trained only on L-band signatures; frequency-agile X-band radars need retraining." Panels respect honesty over false confidence.

Assessment Rubric — Five Criteria, Each Marked Out of 20

Criterion	Weight	Distinction (18–20)	Merit (14–17)	Pass (10–13)	Fail (<10)
Problem definition & context	20%	precisely quantified; operationally grounded; cost of inaction	clear, most context	lacks operational grounding	vague, generic
Algorithm justification	25%	compared with 2+ alternatives; cites course metrics	one alternative	no clear justification	wrong problem type
Results & validation	25%	per-class metrics, CIs, method, baseline	several metrics	primary metric only	no context or baseline
Operational applicability	20%	autonomy, latency, ROE, sim-to-real gap, plan	most factors	gaps remain	purely academic
Critical evaluation & ethics	10%	limitations, IHL, adversarial robustness, failure modes	two limitations	one limitation	none



Criterion weights (%).

Peer evaluation contributes 20% of the project grade: each student rates OTHER groups on a 5-point scale for clarity, technical depth, operational relevance and question handling; moderated by the instructor.

Project Topic Catalogue — Eight Suggested Themes

T1 Threat classification (L2). IADS classifier for a new sensor (EO/IR, SAR, SIGINT); compare RF, XGBoost, SVM; fix imbalance; pick a platform. **Benchmark:** RF 97.5%.

T2 Mission-planning regression (L3). Predict fuel, duration, munitions or rescue time from JANUS data; OLS vs Ridge vs XGBoost; fuse $\geq$2 inputs. **Benchmark:** R^2 0.912.

T3 OPFOR pattern intelligence (L4). Cluster movement/comms/logistics data with k-Means + DBSCAN + PCA; find one unexpected cluster. **Benchmark:** silhouette 0.712.

T4 Cyber-defence anomaly (L5). Tiered IF + AE for a military network; test on one novel attack; compute daily load. **Benchmark:** AE TPR 95.8%.

T5 Wargame decision support (L6). DQN advisor with shaping and curriculum; win rate vs random and vs human; Q-value output. **Benchmark:** 93% vs random.

T6 Autonomous platform control (L7). PPO or SAC for a drone, interceptor, ship autopilot or submarine evasion; domain randomisation. **Benchmark:** SAC 96.2%.

T7 Multi-agent operations (L8). MARL for 3+ agents with CTDE (QMIX or MAPPO); identify an emergent behaviour. **Benchmark:** CommsMAPPO 88.1%.

T8 Original proposal. Combine 2+ lectures; approved by Session 8; novel data or scenario, clear operational context, one course baseline.

The Q&A Panel — It Tests Judgement, Not Algorithm Recall

Every question probes one of three things

① **Depth:** "You reported 94% accuracy. What is the AUROC? The confusion matrix for the rarest class?" ② **Judgement:** "Your DQN won 87% in simulation. Command is considering Level-3 deployment. Your recommendation?" ③ **Intellectual honesty:** "Give me a specific scenario where your model would fail."

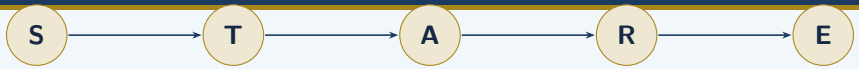
DEPTH | JUDGEMENT | HONESTY

beyond the headline | real-world implications | failure modes

Anticipated questions by topic

Data (L1): "How did you rule out survivorship bias?" "What if the enemy changes behaviour to fool your classifier?" **Performance (L2–3):** "What is the operational cost of a false positive?" "Show me per-class recall." **Clustering/anomaly (L4–5):** "How do you know the clusters are real? ARI?" "2% FPR on 1M packets/day is 20,000 false alerts — how does an analyst cope?" **RL/MARL (L6–8):** "What is the sim-to-real gap?" "Who is accountable if the agent errs?"

Handling Tough Questions — the STARE Framework



Worked STARE answer

Q: "Is your detector robust to adversarial inputs?"
S: "Validated on 100,000 packets: TPR 95%, FPR 0.8%." **T:** "Slide 4 shows the per-attack breakdown." **A:** "We did not test adversarial perturbations." **R:** "Autoencoders can be evaded by attacks that mimic normal reconstruction, as R2L partly does." **E:** "Next: red-team with crafted traffic and add an Isolation-Forest cross-check."

Five phrases that impress panels

1. "We benchmarked against Lecture N's [X%] and achieved [Y%]: a [Z]-point difference due to [reason]."
2. "We chose [A] over [B] because our data had [property] that favours [A]."
3. "We estimate a sim-to-real gap of $\sim$[X]% from L7 (PPO 94% $\rightarrow$ 78%) and designed domain randomisation to close it."
4. "Our rarest class has recall [R]; we set the threshold for the cost of a miss."
5. "We recommend Level 2: the system advises, the commander decides."

Common Project Pitfalls — Avoid These in Your Presentation

1 Accuracy-only reporting. 94% without confusion matrix, per-class recall or AUC is misleading on imbalanced data.	**2 No baseline.** Compare with the majority class, the course benchmark and an alternative algorithm — 87% means nothing alone.	**3 Ignoring data quality.** Every real dataset has bias, shift or label noise; skipping L1 is a red flag.	**4 Simulation = readiness.** "Ready to deploy" from simulation alone; the sim-to-real gap is never zero.
5 Missing ethics. No autonomy level, ROE, IHL, accountability chain or failure consequences.	**6 Wrong algorithm.** Deep learning by default when a simpler model works; supervised methods with no labels.	**7 Misread statistics.** Adding percentages, ignoring base rates, "not significant = no effect", correlation as cause (L1 Section 0).	**8 No uncertainty.** Point results without confidence intervals or repeated runs; a 1-point lead may be noise.

Pitfalls 1–6 are from the original lecture pack; 7–8 add the Lecture 1 data-interpretation lessons.

Presenting Evidence Responsibly — Lessons From Other Disciplines

MEDICINE
Clinical trials report to the CONSORT standard: how many enrolled, how many dropped out, the pre-specified outcome, effect size with confidence interval.
Your project: report N, exclusions, the primary metric fixed in advance, and CIs.

AERONAUTICS
Aviation's checklist and crew-resource-management culture lets any crew member challenge the captain — decisions are verified, not assumed.
Your project: a pre-deployment checklist and a clear human override.

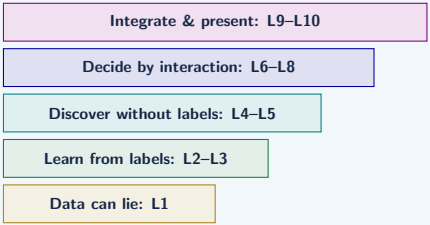
ENGINEERING
Design reviews use failure-mode and effects analysis (FMEA): list how the system can fail, how likely, how severe, and how it is detected.
Your project: a failure-mode table for your model.

PSYCHOLOGY
After the replication crisis, pre-registration — publishing hypotheses and analysis plans before seeing the data — became good practice.
Your project: state your hypothesis and metric before tuning.

STATISTICS
Reproducibility: fixed random seeds, shared code and data, repeated runs with variance reported.
Your project: report mean $\pm$ standard deviation over at least 5 seeds for RL results.

MILITARY PRACTICE
Red teaming and after-action reviews: an independent team attacks the plan; lessons are recorded without blame.
Your project: let another group red-team your model before the panel does.

Course Conclusion — The Arc of Ten Lectures



The habit that matters most

Ask of every dataset and every AI output: *Who collected this? Under what conditions? What is missing? What changed between collection and deployment? How would I know if it were wrong?* That question will save lives.

What you can now do: frame defence problems as ML tasks; choose and justify algorithms with numbers; validate honestly; interpret results without the classic statistical errors; design human–machine teaming that keeps humans accountable.

AI will not replace military commanders — but commanders who understand AI's strengths *and its failure modes* will lead those who do not.

Recommended Reading — Going Further

Core texts

- S. Russell & P. Norvig, *Artificial Intelligence: A Modern Approach* (4th ed., 2020) — the AI foundations of L1
- T. Hastie, R. Tibshirani & J. Friedman, *The Elements of Statistical Learning* (2nd ed., 2009) — L2–L4 in depth
- C. Bishop, *Pattern Recognition and Machine Learning* (2006) — probabilistic view; GMM and EM
- I. Goodfellow, Y. Bengio & A. Courville, *Deep Learning* (2016) — neural networks and autoencoders
- R. Sutton & A. Barto, *Reinforcement Learning: An Introduction* (2nd ed., 2018) — L6–L8

Data interpretation and defence context

- D. Huff, *How to Lie with Statistics* (1954) — the classic on misleading data
- D. Kahneman, *Thinking, Fast and Slow* (2011) — regression to the mean, base rates, judgement biases
- J. Pearl & D. Mackenzie, *The Book of Why* (2018) — correlation, causation and confounding
- P. Scharre, *Army of None* (2018) — autonomous weapons and human control

The companion reference text for each lecture expands every slide of this course in full detail.